

Normal Distribution

Question 1. Tool workers are subject to work-related injuries. One disorder, caused by strains to the hands and wrists, is called carpal tunnel syndrome. It strikes as many as 23,000 workers per year. The U.S. Labor Department estimates that the average cost of this disorder to employers and insurers is approximately \$30,000 per injured worker. Suppose these costs are normally distributed, with a standard deviation of \$9,000.

- a. What proportion of the costs are between \$15,000 and \$45,000?
- b. What proportion of the costs are greater than \$50,000?
- c. What proportion of the costs are between \$5,000 and \$20,000?
- d. Suppose the standard deviation is unknown, but 90.82% of the costs are more than \$7,000. What would be the value of the standard deviation?
- e. Suppose the mean value is unknown, but the standard deviation is still \$9,000. How much would the average cost be if 79.95% of the costs were less than \$33,000?

Solution: Given that, Mean cost, $\mu = \$30,000$; Standard deviation $\sigma = \$9,000$

Let X be a random variable representing the costs. Therefore, X follows a normal distribution with mean, $\mu = \$30,000$ and standard deviation $\sigma = \$9,000$.

- a. Proportion of the costs are between \$15,000 and \$45,000:

$$\begin{aligned} P(15000 < X < 45000) &= P\left(\frac{15000 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{45000 - \mu}{\sigma}\right) \\ &= P\left(\frac{15000 - 30000}{9000} < Z < \frac{45000 - 30000}{9000}\right); \left[\because X \sim N(\mu, \sigma^2), Z = \frac{X - \mu}{\sigma} \sim N(0,1)\right] \\ &= P(-1.67 < Z < 1.67) \\ &= P(Z < 1.67) - P(Z < -1.6) \\ &= 0.9525 - 0.0475 = 0.9050; \text{ [Using standard normal distribution tables]} \end{aligned}$$

So, approximately 90.5% of the costs are between \$15,000 and \$45,000.

- b. Proportion of the costs are greater than \$50,000:

$$\begin{aligned} P(X > 50000) &= P\left(\frac{X - \mu}{\sigma} > \frac{50000 - \mu}{\sigma}\right) \\ &= P\left(Z > \frac{50000 - 30000}{9000}\right) \\ &= P(Z > 2.22) \\ &= 1 - P(Z < 2.22) \\ &= 1 - 0.9868 = 0.0132 \end{aligned}$$

So, approximately 1.32% of the costs are greater than \$50,000.

- c. Proportion of the costs those are between \$5,000 and \$20,000:

$$\begin{aligned}
 P(5000 < X < 20000) &= P\left(\frac{5000 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{20000 - \mu}{\sigma}\right) \\
 &= P\left(\frac{5000 - 30000}{9000} < Z < \frac{20000 - 30000}{9000}\right) \\
 &= P(-2.78 < Z < -1.11) \\
 &= P(Z < -1.11) - P(Z < -2.78) \\
 &= 0.1335 - 0.0027 = 0.1308
 \end{aligned}$$

So, approximately 13.08% of the costs are between \$5,000 and \$20,000.

- d. Suppose the standard deviation is unknown, but 90.82% of the costs are more than \$7,000. What would be the value of the standard deviation?

We are given $P(X > 7,000) = 0.9082$, which means

$$P(X \leq 7,000) = 1 - 0.9082 = 0.0918$$

$$\text{or, } P\left(\frac{X - \mu}{\sigma} \leq \frac{7,000 - \mu}{\sigma}\right) = 0.0918$$

$$\text{or, } P\left(Z \leq \frac{7,000 - 30,000}{\sigma}\right) = 0.0918$$

Find the z-score corresponding to 0.0918 in the standard normal distribution, which is approximately -1.33 .

Therefore,

$$\begin{aligned}
 \frac{7,000 - 30,000}{\sigma} &= -1.33 \\
 \text{or, } \sigma &= \frac{30,000 - 7,000}{1.33} = \frac{23,000}{1.33} \approx 17,293.23
 \end{aligned}$$

So, the standard deviation would be approximately \$17,293.23.

- e. Suppose the mean value is unknown, but the standard deviation is still \$9,000. How much would the average cost be if 79.95% of the costs were less than \$33,000?

We are given,

$$\begin{aligned}
 P(X \leq 33,000) &= 0.7995 \\
 \text{or, } P\left(\frac{X - \mu}{\sigma} \leq \frac{33,000 - \mu}{\sigma}\right) &= 0.7995
 \end{aligned}$$

$$\text{or, } P\left(Z \leq \frac{33,000 - \mu}{9,000}\right) = 0.7995$$

Find the z-score corresponding to 0.7995 in the standard normal distribution, which is approximately 0.84. Therefore,

$$\begin{aligned}
 \frac{33,000 - \mu}{9,000} &= 0.84 \\
 \text{or, } \mu &= 33,000 - (0.84 \times 9,000) = 33,000 - 7,560 = 25,440
 \end{aligned}$$

So, the average cost would be approximately \$25,440.

Question 2. A bank is analyzing the transaction amounts of its customers. The transactions are normally distributed with a mean of 70 and a standard deviation of 3.

- What is the probability that a randomly selected transaction is more than \$75?
- What is the probability that a randomly selected transaction is between \$68 and \$73?
What amount corresponds to the top 10% of all transactions?
- If we randomly sample 25 customers, what is the probability that their average transactions will be greater than \$71?

Solution:

- Probability that a randomly selected transaction is more than \$75:

$$Z = \frac{75 - 70}{3} \approx 1.67$$

Using Z-tables,

$$P(Z > 1.67) \approx 1 - 0.9525 = 0.0475$$

- Percentage of transaction between \$68 and \$73:

$$Z_1 = \frac{68 - 70}{3} = -0.67, \quad Z_2 = \frac{73 - 70}{3} = 1$$

Using Z-tables:

$$P(68 < X < 73) = P(Z < 1) - P(Z < -0.67) \approx 0.8413 - 0.2514 \approx 0.5899$$

Transaction corresponding to the top 10% means 90th percentile: From Z-tables, (Z),

$$X = \mu + Z\sigma = 70 + (1.28)(3) \approx 73.84$$

- Probability that averages transaction of 25 customer is greater than \$71:

$$Z = \frac{71 - 70}{\frac{\sigma}{\sqrt{n}}} = \frac{1}{\frac{3}{\sqrt{25}}} = \frac{1}{0.6} \approx 1.67$$

Using Z-tables: $P(Z > 1.67) \approx 0.0475$

Question 3. In the aftermath of the recent flood in Bangladesh, the depth of water (in inches) measured in 300 different locations was found to be normally distributed with a mean of 68.0 inches and a standard deviation of 3.0 inches. Based on this information:

- How many locations had a water depth of at least 65 inches? [Ans: Number of locations: $0.8413 \times 300 \approx 252$]
- How many locations had a water depth of at most 80 inches? [Ans: 300]
- How many locations had a water depth between 60 inches and 65 inches? [Ans: 46]
- How many locations had a water depth equal to 70 inches? [Ans: 0]

Question 4. If the total cholesterol values BracU faculty members are approximately normally distributed with a mean of 200 mg/100 ml and a standard deviation of 20 mg/100 ml, find the probability that an individual picked at random from this population will have a cholesterol value:

- a. between 180 and 200 mg/100ml. [Ans: 0.3413]
- b. greater than 225 mg/100ml. [Ans: 0.1056]
- c. less than 150 mg/100ml. [Ans: 0.00621]
- d. between 190 and 210 mg/100ml. [Same as (a)]

Question 5. The Phillips Bangladesh manufactures electric bulbs that have a length of life that is normally distributed with mean equal to 800 hours and standard deviation of 40 hours.

- a. Find the probability that a bulb burns between 778 and 834 hours. [Ans: 0.5111]
- b. Find the probability that a bulb burns after 834 hours. [Ans: 0.1977]

Question 6. The monthly salaries of newly graduated students from BRAC University are normally distributed, with a mean of \$250 and a standard deviation of \$6. Based on this information find the probability that a randomly selected graduated will get salary:

- a. At least \$235. [Ans: 0.993]
- b. Between \$255 to \$260. [Ans: 0.16]
- c. At most \$300. [Ans: 1]
- d. Between \$235 to \$280. [Hint: similar to (b)]

Question 7. Assume that, X is defined as the GPA score of students of the department of CSE of BRAC university in their term final examination was found to follow a normal distribution with a mean of 2.1 and a standard deviation of 0.6.

- a. Find the probability that a given student will score at most 2. [Ans: 0.4325]
- b. Find the probability that a given student will score between 2.5 and 3.5. [Ans: 0.2416]
- c. Find the probability that a given student will score at least 3. [Ans: 0.0668]

Question 8. A recent study of the hourly wages of maintenance crews for major airlines showed that the mean hourly salary was 165\$ with standard deviation of 35\$. If we select a crew member at random

- a. What is the probability that the crew member earns between 180\$ to 200\$ per hour? [Ans: 0.1749]
- b. What is the probability that the crew member earns at least 125\$ per hour? [Ans: 0.8729]
- c. In what interval would you expect the central 95% of the value to be found? [Hints: 3sigma rules. Ans: 95, 235]

Question 9. An expert witness in a paternity suit testifies that the length (in days) of human gestation is approximately normally distributed with parameters $\mu = 270$ and $\sigma^2 = 100$. The defendant in the suit is able to prove that he was out of the country during a period that began 290 days before the birth of the child and ended 240 days before the birth. If the defendant was, in fact, the father of the child, what is the probability that the mother could have had the very long or very short gestation indicated by the testimony?

Solution:

Let X denote the length of the gestation, and assume that the defendant is the father. Then the probability that the birth could occur within the indicate period is

$$P(X > 290 \mid X < 240)$$

$$= P(X > 290) + P(X < 240)$$

$$= P\left\{\frac{X - 270}{10} > 2\right\} + P\left\{\frac{X - 270}{10} < -3\right\}$$

$$= 1 - \Phi(2) + 1 - \Phi(3) \approx .0241$$